MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Traceveil

TRACEVEIL

AI-Assisted Real-Time Smart Home IoT Digital Forensics and Incident Response Platform

Group 18

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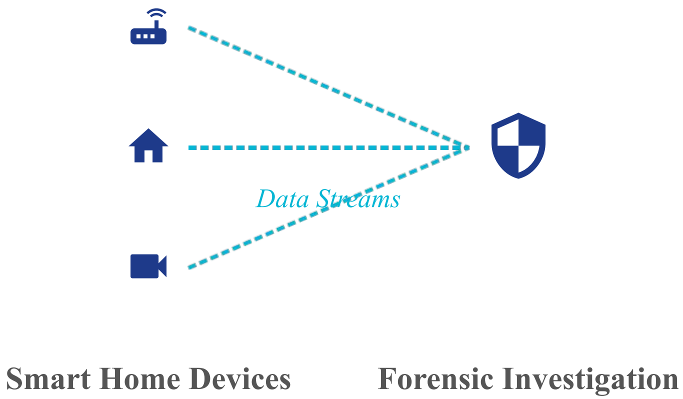

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From Fragmented IoT Activity to Investigable Evidence

Why This Topic?

- Smart-home evidence is fragmented across devices and vendors
- Live alerts alone do not provide complete forensic context
- IoT telemetry is heterogeneous and privacy-sensitive
- Public datasets & live telemetry enable repeatable experiments

The Core Forensic Challenge

Siloed & Fragmented Data

IoT evidence is heavily fragmented across diverse devices, proprietary formats, drifted timestamps, and isolated identifiers.

This makes subsequent forensic timeline reconstruction highly complex and error-prone.

1. Smart Home Devices
Historical Evidence & Live Telemetry

→

2. Fragmented Logs
Siloed formats & drifted timestamps

→

3. Traceveil Engine
Canonical Normalization & Correlation

→

4. Investigation
Correlated Alerts, Timelines, & Evidence

Literature Survey

Research Work	Main Focus	Relevance to Traceveil
■**TON_IoT** — 2020	Heterogeneous IoT/IIoT telemetry dataset	Supports batch evidence ingestion and normalization
■**CICIoT2023** — 2023	Large-scale IoT attack dataset and benchmark	Supports evaluation of detection workflows
■**Smart Home Forensics** — 2020	Multi-device smart-home evidence analysis	Supports cross-device evidence correlation
■**IoT Forensic Challenges & Opportunities** — 2019	IoT evidence acquisition and forensic challenges	Motivates a unified forensic workflow

Supporting Standards: NIST SP 800-86 · NIST SP 800-92

Tool-Based Comparison & Research Positioning

Capability	Defender for IoT	Device Defender	Splunk ES / QRadar	Traceveil (Ours)
Live IoT Monitoring	Enterprise capability	AWS IoT fleet monitoring	Security event monitoring	**Controlled laboratory telemetry**
Historical Evidence	Platform ecosystem support	Audit & monitoring history	Strong log investigation	**Core project focus**
Event Normalization	Platform-specific	Platform-specific	General security schemas	**One canonical IoT event model**
Timeline Reconstruction	Broader workflows	Not primary focus	Investigation capabilities	**Core project focus**
Cross-Device Correlation	Enterprise correlation	Device/fleet-oriented	Broad event correlation	**IoT-focused correlation**
AI Investigation	Platform-specific	Not primary focus	Product-dependent	**Planned local evidence-grounded Qwen assistant**
Primary Scope	Enterprise IoT/OT	AWS IoT environments	Enterprise SOC/SIEM	**Academic smart-home IoT DFIR prototype**

Research Positioning

Traceveil focuses on unifying batch evidence + controlled live telemetry through one canonical event model and one deterministic forensic investigation pipeline.

Objectives & Project Scope

Objectives

- Ingest public datasets & uploaded evidence
- Receive controlled live IoT telemetry
- Validate & normalize accepted records
- Preserve source & device provenance
- Apply rules & behavioral baselines
- Calculate bounded risk scores
- Correlate events & reconstruct timelines
- Generate charts & entity relationships
- Provide AI-assisted investigation

In Scope

- TON_IoT and CICIoT2023
- Simulated forensic cases
- Controlled laboratory IoT telemetry
- Arduino-compatible telemetry source
- React investigation dashboard
- Local Qwen assistance
- SQLite case history
- Investigator-approved alerts & reports

Out of Scope

- Unauthorized device access
- Arbitrary network interception
- Production-scale monitoring
- Autonomous containment
- Attack execution
- Legal-admissibility claims
- Replacement of enterprise platforms

Acquire → Validate → Normalize → Analyse → Preserve → Deliver → Investigate

Proposed System Architecture

Batch Evidence
- TON_IoT
- CICIoT2023
- Uploaded files
- Simulated cases

Controlled Live IoT
- Arduino telemetry
- IoT test endpoint

Validation & Normalization

Canonical Event Model

Deterministic DFIR Engine
Detection rulesBehavioural baselinesRisk scoringCorrelationTimeline reconstructionGraph calculations

Python remains the computational authority.

Evidence Store
Secure normalized case history database

FastAPI APIs & Real-Time Delivery
High-performance telemetry broadcast pipeline

Visualization + AI Layer
Local offline AI assistance for investigator queries.

** AI only consumes verified results and does not calculate forensic values.*

React Investigation Dashboard
Interactive interface rendering live charts, comprehensive event timelines, and AI-assisted triage reports.

Technology Stack

Presentation Layer	ReactViteTailwind CSSshadcn/uiRechartsReact Flow
Application & Analysis	PythonFastAPIPydanticPandasNetworkX
Data & AI	SQLiteOllamaQwen 9B Q4_K_M
Comm & Dev	SMTPGitGitHubPytestVitestMQTT/HTTP (Planned)SSE/WebSockets (Planned)
Hardware Layer	Dev LaptopArduino MCUIoT Test DeviceAuthorized Local Network

Expected Deliverables & Demonstration

Expected Deliverables

- Batch evidence ingestion pipeline
- Controlled live telemetry collector
- Canonical IoT event model
- Deterministic detection and risk engine
- Event correlation and incident timelines
- Device/entity graphs and charts
- Investigation dashboard
- Local AI investigation assistant
- Reports and auditable evidence history

End-to-End Demonstration

1. Normal Device Telemetry
2. Controlled Suspicious Scenario
3. Telemetry Received & Validated
4. Deterministic Rule Triggered
5. Live Alert Appears
6. Evidence Preserved
7. Timeline & Device Graph Updated
8. Incident Available for Later Review

Forensic Authority Mandate

Disabling the AI model must not change any forensic result.

Conclusion

Traceveil unifies real-time awareness with forensic reconstruction.

- **Canonical Model:** Historical and live evidence share one canonical event model.
- **Reproducibility:** Deterministic analytics preserve absolute reproducibility.
- **Availability:** Live incidents remain available for later forensic investigation.
- **Clarity:** Evidence correlation supports clearer incident reconstruction.
- **AI Safeguards:** AI explains verified results rather than creating forensic findings.
- **Academic Focus:** Traceveil is designed as a transparent academic DFIR prototype.

Batch Evidence + Live IoT

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Deterministic DFIR

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Explainable Investigation

References & Closing

1. **NIST SP 800-86** — Guide to Integrating Forensic Techniques into Incident Response
2. **NIST SP 800-92** — Guide to Computer Security Log Management
3. **TON_IoT** — IEEE Access, 2020
4. **CICIoT2023** — Sensors, 2023
5. **Microsoft Defender for IoT** documentation
6. **AWS IoT Device Defender** documentation
7. **Splunk Enterprise Security** documentation
8. **IBM QRadar** documentation
9. **Ollama Structured Outputs** documentation

THANK YOU

Questions?